

## Statistics for Data Science - 2

### Week 5 graded Assignment

#### Solution

1. A person randomly chooses a battery from a store which has 40 batteries of type A and 60 batteries of type B. Battery life of type A and type B batteries are exponentially distributed with average life of 4 years and 6 years, respectively. If the chosen battery lasts for 5 years, what is the probability that the battery is of type A?

(a)  $\frac{1}{1 + e^{\frac{5}{12}}}$

(b)  $\frac{1}{1 + e^{\frac{-5}{12}}}$

(c)  $\frac{e^{\frac{-4}{5}}}{1 + e^{\frac{-6}{5}}}$

(d)  $\frac{e^{\frac{-6}{5}}}{1 + e^{\frac{-4}{5}}}$

**Solution:**

Define a event  $X$  as follows:

$$X = \begin{cases} 1 & \text{If the chosen battery is of type A} \\ 0 & \text{If the chosen battery is of type B} \end{cases}$$

Let  $Y$  denote the battery life of the chosen battery.

By the given information, we have

$Y|X = 1 \sim \text{Exp}(\frac{1}{4})$  and

$Y|X = 0 \sim \text{Exp}(\frac{1}{6})$

It implies that

$$f_{Y|X=1}(y) = \frac{1}{4}e^{\frac{-y}{4}}; y > 0 \text{ and}$$

$$f_{Y|X=0}(y) = \frac{1}{6}e^{\frac{-y}{6}}; y > 0$$

Also given that

$$P(X = 1) = \frac{40}{100} = \frac{2}{5} \text{ and}$$

$$P(X = 0) = \frac{60}{100} = \frac{3}{5}$$

To find:  $f_{X|Y=5}(1)$ . Now,

$$\begin{aligned} f_{X|Y=5}(1) &= \frac{f_{Y|X=1}(5) \cdot P(X = 1)}{f_Y(5)} \\ &= \frac{f_{Y|X=1}(5) \cdot P(X = 1)}{f_{Y|X=1}(5) \cdot P(X = 1) + f_{Y|X=0}(5) \cdot P(X = 0)} \\ &= \frac{\frac{1}{4} e^{-\frac{5}{4}} \cdot \frac{2}{5}}{\frac{1}{4} e^{-\frac{5}{4}} \cdot \frac{2}{5} + \frac{1}{6} e^{-\frac{5}{6}} \cdot \frac{3}{5}} \\ &= \frac{\frac{1}{10} e^{-\frac{5}{4}}}{\frac{1}{10} e^{-\frac{5}{4}} + \frac{1}{10} e^{-\frac{5}{6}}} \\ &= \frac{e^{-\frac{5}{4}}}{e^{-\frac{5}{4}} + e^{-\frac{5}{6}}} \\ &= \frac{1}{1 + e^{\frac{5}{12}}} \end{aligned}$$

2. Let  $Y = XZ + X$ , where  $X \sim \text{Uniform}\{1, 2, 3\}$  and  $Z \sim \text{Normal}(1, 4)$  are independent. Find the value of  $f_{X|Y=2}(2)$ .

- (a)  $\frac{3 \exp(\frac{1}{8})}{3 \exp(\frac{1}{8}) + 6 + 2 \exp(\frac{2}{9})}$   
 (b)  $\frac{3 \exp(\frac{-1}{8})}{3 \exp(\frac{-1}{8}) + 6 + 2 \exp(\frac{-2}{9})}$   
 (c)  $\frac{2 \exp(\frac{-2}{9})}{3 \exp(\frac{-1}{8}) + 6 + 2 \exp(\frac{-2}{9})}$   
 (d)  $\frac{6}{3 \exp(\frac{-1}{32}) + 6 + 2 \exp(\frac{-1}{18})}$

**Solution:**

Given that  $X \sim \text{Uniform}\{1, 2, 3\}$  and  $Z \sim \text{Normal}(1, 4)$  are independent.

$$Y = XZ + X$$

It implies that

$$\begin{aligned}
Y|X=1 &= Z+1 \sim \text{Normal}(2, 4) \\
Y|X=2 &= 2Z+2 \sim \text{Normal}(4, 16) \\
Y|X=3 &= 3Z+3 \sim \text{Normal}(6, 36)
\end{aligned}$$

Therefore,

$$\begin{aligned}
f_{Y|X=1}(y) &= \frac{1}{2\sqrt{2\pi}} \exp\left(\frac{-(y-2)^2}{8}\right) \\
f_{Y|X=2}(y) &= \frac{1}{4\sqrt{2\pi}} \exp\left(\frac{-(y-4)^2}{32}\right) \\
f_{Y|X=3}(y) &= \frac{1}{6\sqrt{2\pi}} \exp\left(\frac{-(y-6)^2}{72}\right)
\end{aligned}$$

To find:  $f_{X|Y=2}(2)$ .

$$\begin{aligned}
f_{X|Y=2}(2) &= \frac{f_{Y|X=2}(2) \cdot f_X(2)}{f_{Y|X=2}(2) \cdot f_X(2) + f_{Y|X=1}(2) \cdot f_X(1) + f_{Y|X=3}(2) \cdot f_X(3)} \\
&= \frac{\frac{1}{4\sqrt{2\pi}} \exp\left(\frac{-(2-4)^2}{32}\right) \cdot \frac{1}{3}}{\frac{1}{4\sqrt{2\pi}} \exp\left(\frac{-(2-4)^2}{32}\right) \cdot \frac{1}{3} + \frac{1}{2\sqrt{2\pi}} \exp\left(\frac{-(2-2)^2}{8}\right) \cdot \frac{1}{3} + \frac{1}{6\sqrt{2\pi}} \exp\left(\frac{-(2-6)^2}{72}\right) \cdot \frac{1}{3}} \\
&= \frac{\frac{1}{4} \exp\left(\frac{-1}{8}\right)}{\frac{1}{4} \exp\left(\frac{-1}{8}\right) + \frac{1}{2} \exp(0) + \frac{1}{6} \exp\left(\frac{-2}{9}\right)} \\
&= \frac{3 \exp\left(\frac{-1}{8}\right)}{3 \exp\left(\frac{-1}{8}\right) + 6 + 2 \exp\left(\frac{-2}{9}\right)}
\end{aligned}$$

3. The joint pdf of two continuous random variables  $X$  and  $Y$  is given by

$$f_{XY}(x, y) = \begin{cases} 4xy & 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Are  $X$  and  $Y$  independent?

1. Yes
2. No

**Solution:**

First we will calculate the marginal densities of  $X$  and  $Y$ .

For  $0 \leq x \leq 1$

$$\begin{aligned} f_X(x) &= \int_0^1 f_{XY}(x, y) dy \\ &= \int_0^1 4xy dy \\ &= 2xy^2 \Big|_0^1 \\ &= 2x \end{aligned}$$

For  $0 \leq y \leq 1$

$$\begin{aligned} f_Y(y) &= \int_0^1 f_{XY}(x, y) dx \\ &= \int_0^1 4xy dx \\ &= 2x^2 y \Big|_0^1 \\ &= 2y \end{aligned}$$

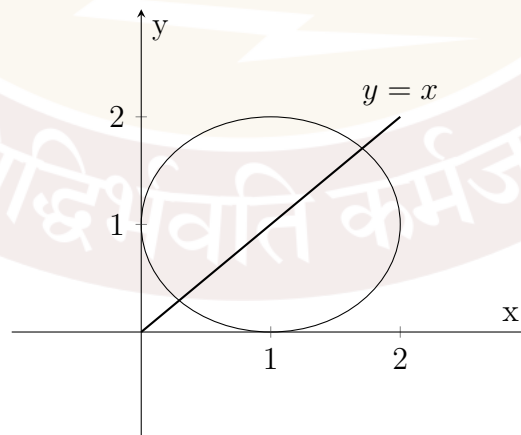
Therefore,

$$f_X(x) \cdot f_Y(y) = 4xy = f_{XY}(x, y)$$

It implies that  $X$  and  $Y$  are independent random variables.

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4. Let  $(X, Y) \sim \text{Uniform}(D)$ , where  $D = \{(x, y) : (x - k)^2 + (y - k)^2 \leq r\}$ . Calculate  $P(X \geq Y)$ .

**Solution:**



The region  $X \geq Y$  will be the lower half part of the circle.

Therefore,

$$\begin{aligned} P(X \geq Y) &= \frac{\text{Area of lower half circle}}{\text{Area of the circle}} \\ &= \frac{\pi(1)^2/2}{\pi(1)^2} \\ &= \frac{1}{2} \end{aligned}$$

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5. Let  $(X, Y) \sim \text{Uniform}(D)$ , where  $D = \{(x, y) : y \leq 2x, 0 < x < 1, 0 < y < 2\} \cup [1, 2] \times [0, 2]$ . Find the marginal density of  $X$ .

(a)

$$f_X(x) = \begin{cases} \frac{2x}{3} + \frac{2}{3} & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(b)

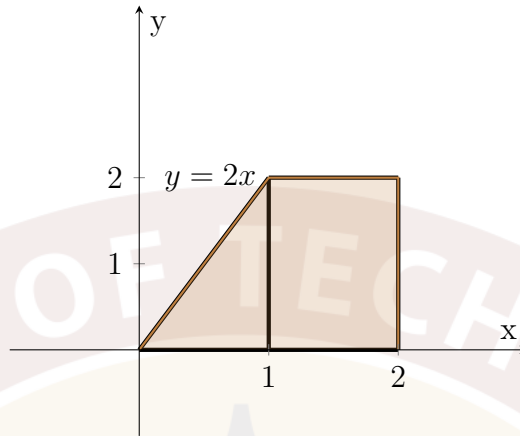
$$f_X(x) = \begin{cases} \frac{2x}{3} + \frac{1}{3} & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(c)

$$f_X(x) = \begin{cases} \frac{2x}{3} & 0 \leq x \leq 1 \\ \frac{2}{3} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

(d)

$$f_X(x) = \begin{cases} \frac{2x}{3} & 0 \leq x \leq 1 \\ \frac{1}{3} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$



$D$  denotes the area of the support  $(X, Y)$ .

Area of  $D = \frac{1}{2} \times 1 \times 2 + 1 \times 2 = 3$

Since  $(X, Y) \sim \text{Uniform}(D)$ , it implies that

$$f_{XY}(x, y) = \frac{1}{3}, \quad x, y \in D$$

We know that  $f_X(x) = \int f_{XY}(x, y) dy$

For  $0 < x < 1$

$$\begin{aligned} f_X(x) &= \int_0^{2x} \frac{1}{3} dy \\ &= \frac{1}{3} y \Big|_0^{2x} \\ &= \frac{2x}{3} \end{aligned}$$

For  $1 < x < 2$

$$\begin{aligned} f_X(x) &= \int_0^2 \frac{1}{3} dy \\ &= \frac{1}{3} y \Big|_0^2 \\ &= \frac{2}{3} \end{aligned}$$

Therefore, marginal density of  $X$  is given by

$$f_X(x) = \begin{cases} \frac{2x}{3} & 0 \leq x \leq 1 \\ \frac{2}{3} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

6. The joint pdf of two random variables  $X$  and  $Y$  is given by

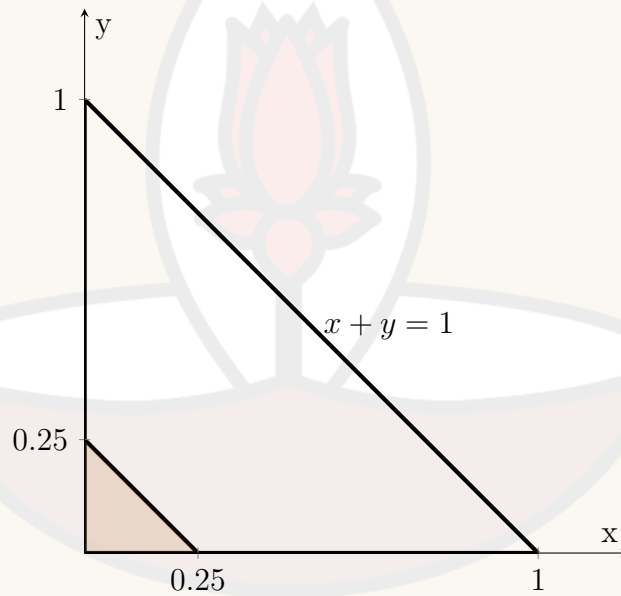
$$f_{XY}(x, y) = \begin{cases} 24xy & 0 \leq x \leq 1, 0 \leq y \leq 1, x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Choose the correct option(s).

- (a)  $P(X + Y \leq \frac{1}{4}) = \frac{1}{2}$
- (b)  $P(X + Y \leq \frac{1}{2}) = \frac{1}{16}$
- (c)  $X$  and  $Y$  are independent random variables.
- (d)  $X$  and  $Y$  are dependent random variables.

**Solution:**

Option (a)



Orange region will denote  $X + Y \leq \frac{1}{4}$ . Now,

$$P(X + Y \leq \frac{1}{4}) = \int_{y=0}^{1/4} \int_{x=0}^{1/4-y} f_{XY}(x, y) dx dy$$

$$= \int_{y=0}^{1/4} \int_{x=0}^{1/4-y} 24xy dx dy$$

$$= \int_{y=0}^{1/4} 12x^2 y \Big|_{x=0}^{1/4-y} dy$$

$$= \int_{y=0}^{1/4} 12y \left( \frac{1}{4} - y \right)^2 dy$$

$$= \int_{y=0}^{1/4} \frac{12}{16} y (1 - 4y)^2 dy$$

$$= \frac{3}{4} \int_{y=0}^{1/4} y (1 + 16y^2 - 8y) dy$$

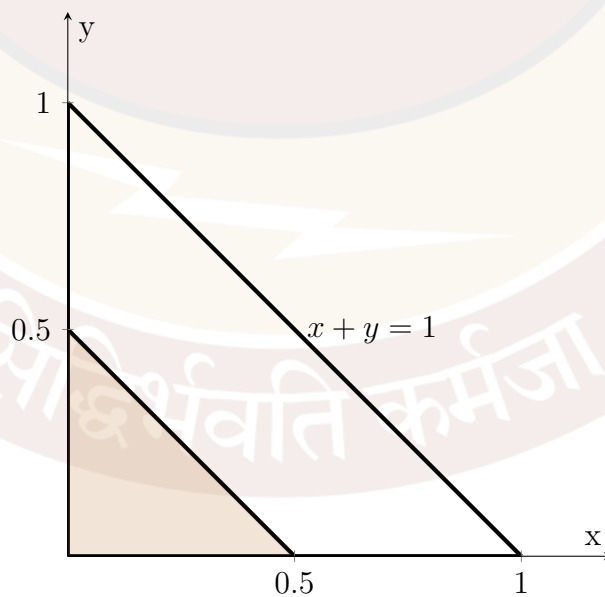
$$= \frac{3}{4} \left( \frac{y^2}{2} + 4y^4 - \frac{8y^3}{3} \right) \Big|_{y=0}^{1/4}$$

$$= \frac{3}{4} \left( \frac{1}{32} + \frac{1}{64} - \frac{1}{24} \right)$$

$$= \frac{3}{4} \cdot \frac{1}{192} = \frac{1}{256}$$

Hence, option (a) is wrong.

Option (b)

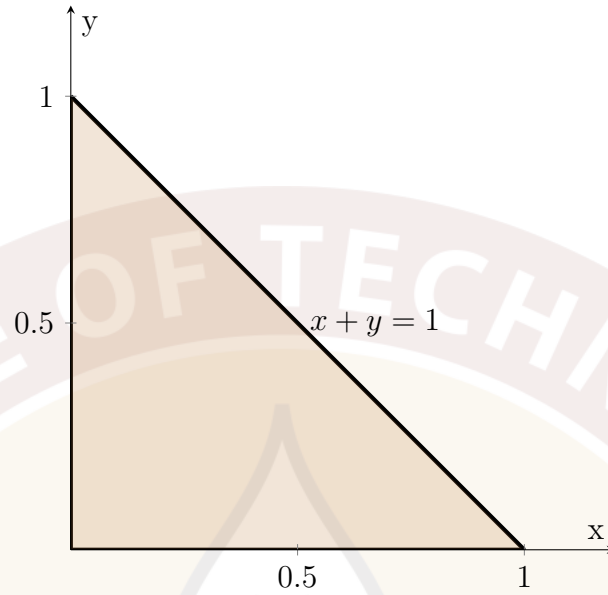


Orange region will denote  $X + Y \leq \frac{1}{2}$ . Now,

$$\begin{aligned}
 P(X + Y \leq \frac{1}{2}) &= \int_{y=0}^{1/2} \int_{x=0}^{1/2-y} f_{XY}(x, y) dx dy \\
 &= \int_{y=0}^{1/2} \int_{x=0}^{1/2-y} 24xy dx dy \\
 &= \int_{y=0}^{1/2} 12x^2 y \Big|_{x=0}^{1/2-y} dy \\
 &= \int_{y=0}^{1/2} 12y \left( \frac{1}{2} - y \right)^2 dy \\
 &= \int_{y=0}^{1/2} \frac{12}{4} y(1 - 2y)^2 dy \\
 &= 3 \int_{y=0}^{1/2} y(1 + 4y^2 - 4y) dy \\
 &= 3 \left( \frac{y^2}{2} + y^4 - \frac{4y^3}{3} \right) \Big|_{y=0}^{1/2} \\
 &= 3 \left( \frac{1}{8} + \frac{1}{16} - \frac{1}{6} \right) \\
 &= 3 \times \frac{2}{96} = \frac{1}{16}
 \end{aligned}$$

Hence, option (b) is correct.

Option (c) and (d)



For  $0 < x < 1$

$$\begin{aligned}
 f_X(x) &= \int_{y=0}^{1-x} f_{XY}(x, y) dy \\
 &= \int_{y=0}^{1-x} 24xy dy \\
 &= 12xy^2 \Big|_{y=0}^{1-x} \\
 &= 12x(1-x)^2
 \end{aligned}$$

For  $0 < y < 1$

$$\begin{aligned}
 f_Y(y) &= \int_{x=0}^{1-y} f_{XY}(x, y) dx \\
 &= \int_0^{1-y} 24xy dx \\
 &= 12x^2y \Big|_{x=0}^{1-y} \\
 &= 12y(1-y)^2
 \end{aligned}$$

Therefore,  $f_X(x) \cdot f_Y(y) = 144xy(1-x)^2(1-y)^2 \neq f_{XY}(x, y)$

Hence,  $X$  and  $Y$  are not independent.

7. The joint pdf of two random variables  $X$  and  $Y$  is given by

$$f_{XY}(x, y) = \begin{cases} 3xy(1-x) & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Calculate  $P(X > \frac{1}{2} | Y = 1)$ .

**Solution:**

We know that

$$P(a < X < b | Y = y) = \frac{f_{XY}(a < X < b, y)}{f_Y(y)}$$

Now,

$$\begin{aligned} f_Y(y) &= \int_0^1 3xy(1-x)dx \\ &= \int_0^1 (3xy - 3x^2y)dx \\ &= \left( \frac{3x^2y}{2} - x^3y \right) \Big|_0^1 \\ &= \frac{3y}{2} - y = \frac{y}{2} \end{aligned}$$

Therefore,  $f_Y(1) = \frac{1}{2}$

Now,

$$\begin{aligned} P(X > \frac{1}{2} | Y = 1) &= \frac{f_{XY}(X > \frac{1}{2}, Y = 1)}{f_Y(1)} \\ &= 2f_{XY}(X > \frac{1}{2}, Y = 1) \\ &= \int_{x=\frac{1}{2}}^1 2(3x(1-x))dx \\ &= 6 \int_{\frac{1}{2}}^1 (x - x^2)dx \\ &= 6 \left( \frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_{\frac{1}{2}}^1 \\ &= 6 \left( \frac{1}{2} - \frac{1}{3} \right) - 6 \left( \frac{1}{8} - \frac{1}{24} \right) = 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

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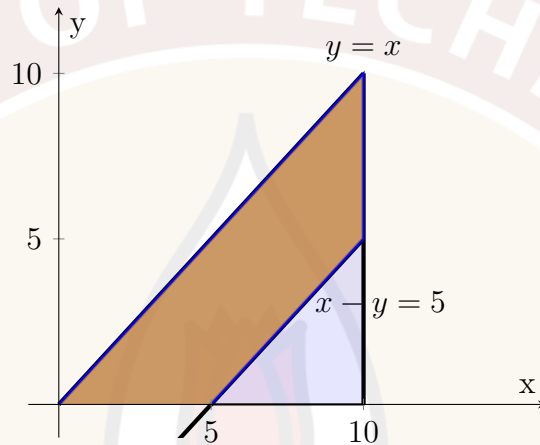
8. The amount of milk (in litres) in a shop at the beginning of any day is a random amount  $X$  from which a random amount  $Y$  (in litres) is sold during that day. Assume that the

joint density function of  $X$  and  $Y$  is given by

$$f_{XY}(x, y) = \begin{cases} \frac{1}{50} & 0 \leq x \leq 10, 0 \leq y \leq x \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that amount of milk left at the end of day is less than 5 litres. Write your answer correct to two decimal points.

**Solution:**



$X$  denotes the amount of milk at the beginning of any day and  $Y$  denotes the amount of milk which is sold during that day.

Therefore, amount of milk left at the end of the day will be denoted by  $X - Y$ .

To find:  $P(X - Y < 5)$

In the diagram above, brown region denotes  $X - Y < 5$  and brown + blue region denotes the support of  $X$  and  $Y$ .

Area of the support( $X, Y$ ) =  $\frac{1}{2} \times 10 \times 10 = 50$ .

Area of brown region = Area of support( $X, Y$ ) – area of blue region

$$\Rightarrow \text{area of brown region} = 50 - \frac{1}{2} \times 5 \times 5 = \frac{75}{2}$$

Therefore,

$$\begin{aligned} P(X - Y < 5) &= \frac{\text{area of brown region}}{\text{area of support}} \\ &= \frac{75/2}{50} \\ &= \frac{75}{100} \end{aligned}$$

9. The joint pdf of two continuous random variables  $X$  and  $Y$  is given by

$$f_{XY}(x, y) = \begin{cases} ke^{-(x+y)} & x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of  $P(X \geq 5, Y \leq 5)$ .

- (a)  $e^{-10}$
- (b)  $(e^{-5} - 1)e^{-5}$
- (c)  $(1 - e^{-5})e^{-5}$
- (d)  $(e^{-5} + 1)e^{-5}$

**Solution:**

We know that

$$\iint_{\text{Supp}(X,Y)} f_{XY} dx dy = 1$$

Therefore,

$$\begin{aligned} \int_{y=0}^{\infty} \int_{x=0}^{\infty} (ke^{-(x+y)}) dx dy &= 1 \\ \Rightarrow k \int_{y=0}^{\infty} \int_{x=0}^{\infty} e^{-y} e^{-x} dx dy &= 1 \\ \Rightarrow k \int_{y=0}^{\infty} e^{-y} (-e^{-x}) \Big|_0^{\infty} dy &= 1 \\ \Rightarrow k \int_{y=0}^{\infty} e^{-y} (0 + 1) dy &= 1 \\ \Rightarrow k \int_{y=0}^{\infty} e^{-y} dy &= 1 \\ \Rightarrow k (-e^{-y}) \Big|_0^{\infty} &= 1 \\ \Rightarrow k (0 + 1) &= 1 \\ \Rightarrow k &= 1 \end{aligned}$$

To find:  $P(X \geq 5, Y \leq 5)$

Now,

$$\begin{aligned} P(X \geq 5, Y \leq 5) &= \int_{y=0}^5 \int_{x=5}^{\infty} (e^{-(x+y)}) dx dy \\ &= \int_{y=0}^5 \int_{x=5}^{\infty} e^{-y} e^{-x} dx dy \\ &= \int_{y=0}^5 e^{-y} (-e^{-x}) \Big|_5^{\infty} dy \\ &= \int_{y=0}^5 e^{-y} (0 + e^{-5}) dy \\ &= (e^{-5}) \int_{y=0}^5 e^{-y} dy \\ &= (e^{-5}) (-e^{-y}) \Big|_0^5 \\ &= (e^{-5}) (-e^{-5} + 1) \\ &= (e^{-5}) (1 - e^{-5}) \end{aligned}$$

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10. The joint pdf of two random variables  $X$  and  $Y$  is given by

$$f_{XY}(x, y) = \begin{cases} \frac{1}{8}(x + y) & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of  $P\left(\frac{1}{2} \leq y \leq 1 \mid \left(X = \frac{1}{2}\right)\right)$ . Write your answer correct to two decimal points.

**Solution:**

We know that

$$P(a < Y < b \mid X = x) = \frac{f_{XY}(X = x, a < Y < b)}{f_X(x)}$$

Now,

$$\begin{aligned} f_X(x) &= \int_0^2 \frac{1}{8}(x + y) dy \\ &= \frac{1}{8} \left( xy + \frac{y^2}{2} \right) \Big|_0^2 \\ &= \frac{2x + 2}{8} = \frac{x + 1}{4} \end{aligned}$$

Therefore,  $f_X\left(\frac{1}{2}\right) = \frac{3}{8}$

Now,

$$\begin{aligned} P\left(\frac{1}{2} \leq Y \leq 1 \mid X = \frac{1}{2}\right) &= \frac{f_{XY}(X = \frac{1}{2}, \frac{1}{2} \leq Y \leq 1)}{f_X(\frac{1}{2})} \\ &= \int_{1/2}^1 \frac{8}{3} \left[ \frac{1}{8} \left( \frac{1}{2} + y \right) \right] dy \\ &= \int_{1/2}^1 \frac{1}{3} \left( \frac{1}{2} + y \right) dy \\ &= \left( \frac{y}{6} + \frac{y^2}{6} \right) \Big|_{1/2}^1 \\ &= \left( \frac{1}{6} + \frac{1}{6} \right) - \left( \frac{1}{12} + \frac{1}{24} \right) \\ &= \frac{1}{3} - \frac{1}{8} = \frac{5}{24} = 0.20 \end{aligned}$$

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